

KEM PORT, Russian Federation

WMO: 225200

Lat: **64.983N**Lon: **34.800E**Elev: **25**StdP: **14.68**Time Zone: **3.00 (E03)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-7.5	-18.0	2.1	-12.9	-11.7	2.9	-7.6	23.7	21.3	21.9	20.7	3.7	260	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	75.5	63.2	71.4	61.6	68.3	60.2	65.3	72.4	63.4	69.4	61.4	66.9	8.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.5	84.7	68.5	60.6	79.1	66.6	58.7	73.8	64.8	30.2	72.9	28.8	69.3	27.4	67.0	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.3	16.3	DB	-17.3	81.7	6.5	4.6	-21.9	85.0	-25.7	87.7	-29.3	90.3	-34.0	93.7	
			WB	-17.6	67.7	6.4	2.3	-22.2	69.4	-25.9	70.7	-29.5	72.0	-34.1	73.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	15.3	15.9	23.7	33.2	43.5	51.8	58.6	57.0	49.5	38.1	29.2	22.6	
	DBStd	17.01	11.75	11.92	9.33	6.60	7.79	6.79	6.45	6.35	5.74	5.94	6.54	10.46	
	HDD50	5580	1077	954	817	505	232	58	7	10	78	369	624	849	
	HDD65	10381	1542	1374	1282	955	666	399	215	260	466	834	1074	1314	
	CDD50	709	0	0	0	0	31	112	274	228	63	1	0	0	
	CDD65	33	0	0	0	0	0	3	18	12	0	0	0	0	
	CDH74	184	0	0	0	0	12	21	97	54	0	0	0	0	
	CDH80	30	0	0	0	0	2	4	20	4	0	0	0	0	
Wind	WSAvg	8.3	8.6	8.2	8.4	8.1	8.3	7.9	7.1	7.5	8.0	8.9	9.3	9.0	
Precipitation	PrecAvg	17.1	0.8	0.6	0.6	0.9	1.6	2.2	1.8	2.4	2.1	1.9	1.5	1.0	
	PrecMax	27.9	1.7	1.4	1.4	1.9	3.8	4.8	4.3	4.4	4.3	3.3	2.3	2.7	
	PrecMin	10.8	0.2	0.1	0.1	0.0	0.3	0.4	0.1	0.6	0.7	0.2	0.4	0.1	
	PrecStd	3.5	0.4	0.4	0.3	0.5	0.9	1.3	1.1	0.9	1.2	0.8	0.5	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	39.5	45.7	57.6	74.5	77.0	83.0	79.6	68.5	54.3	45.0	40.1	
		MCWB	34.9	36.1	38.1	47.2	55.7	61.8	66.4	65.2	58.9	49.9	43.2	37.6	
	2%	DB	34.3	36.2	40.5	49.2	65.3	70.4	76.0	74.4	64.1	50.5	41.5	37.4	
		MCWB	32.8	34.1	35.1	41.4	51.9	58.0	64.6	62.8	57.4	47.4	40.1	35.4	
	5%	DB	32.4	34.1	37.7	45.4	60.7	65.9	71.7	70.7	60.8	48.1	38.7	35.8	
		MCWB	31.5	32.7	33.7	38.7	50.1	56.2	62.8	61.8	55.1	45.6	37.2	34.1	
	10%	DB	30.3	31.3	35.6	42.4	56.4	62.4	68.6	67.3	58.0	46.0	36.9	34.2	
		MCWB	29.7	30.1	32.7	37.1	48.3	54.8	61.5	60.7	53.8	43.9	35.5	33.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	36.3	38.5	47.8	57.2	65.0	69.1	66.5	60.5	51.1	43.4	38.5
			MCDB	36.0	38.5	44.6	57.2	66.8	74.5	77.3	75.5	65.7	53.4	44.9	39.4
2%		WB	33.1	34.2	35.9	41.7	53.7	60.0	66.2	64.7	58.0	48.3	40.2	35.8	
		MCDB	34.1	35.7	39.4	47.6	63.0	67.0	73.3	72.1	62.5	49.5	41.2	37.0	
5%		WB	31.5	32.8	34.4	39.7	51.1	57.5	64.1	63.0	56.2	45.7	37.4	34.4	
		MCDB	32.3	34.2	37.1	44.2	59.6	63.9	69.6	69.1	60.0	47.4	38.3	35.5	
10%		WB	29.7	30.2	33.1	37.7	48.7	55.8	62.2	61.0	54.4	44.0	35.8	33.0	
		MCDB	30.4	31.3	35.4	41.7	55.7	61.4	67.5	66.3	57.6	46.0	36.7	33.9	
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.2	12.2	11.6	11.8	11.1	10.5	9.7	7.1	6.6	7.9		
		MCDBR	7.9	8.5	11.4	15.6	21.0	18.4	17.5	16.3	14.5	9.6	7.2	7.2	
		MCWBR	7.9	7.8	8.3	9.6	10.9	9.3	8.8	7.6	8.6	7.2	6.7	6.6	
	5% WB	MCDBR	7.4	8.8	10.1	14.4	19.1	14.8	14.7	14.1	12.0	7.9	7.0	6.7	
		MCWBR	7.4	8.2	8.0	9.3	11.7	8.8	8.7	7.9	7.6	7.0	6.7	6.5	
Clear-Sky Solar Irradiance	taub	0.219	0.280	0.309	0.342	0.353	0.349	0.378	0.355	0.331	0.303	0.345	0.356		
	taud	1.957	2.094	2.154	2.275	2.425	2.487	2.436	2.476	2.521	2.363	2.560	2.606		
	Ebn at Noon	89	187	239	262	271	275	261	257	240	186	87	37		
	Edh at Noon	7	19	29	32	31	30	30	27	21	15	7	4		
All-Sky Solar Radiation	RadAvg	39	220	710	1186	1480	1734	1590	1133	641	232	50	11		
	RadStd	4	36	73	113	128	154	174	116	67	30	8	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+1.12	N/A	N/A	N/A	N/A	N/A	N/A	-409	N/A	N/A

Nomenclature: See separate page